TWINGLU

A hybrid digital twin for Type 2 Diabetes

Happiest Health Digital Twin in Healthcare Challenge

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The problem: the doctor sees a snapshot, the disease is a movie

Visits every few months

The doctor sees one moment in time

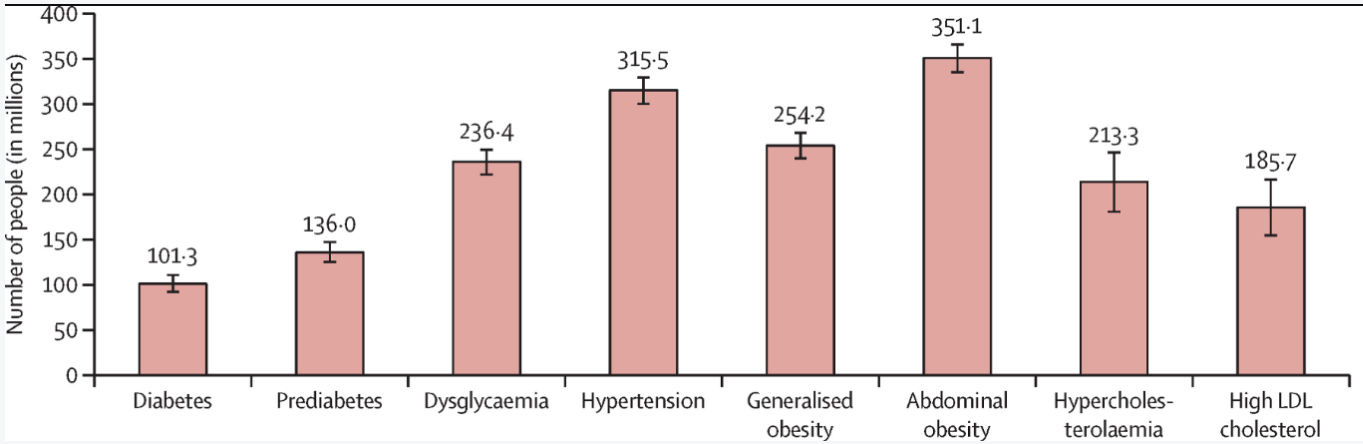
Glucose changes every minute

Meals, walks, sleep and stress all move it

Danger hides in between

Spikes and lows happen between visits

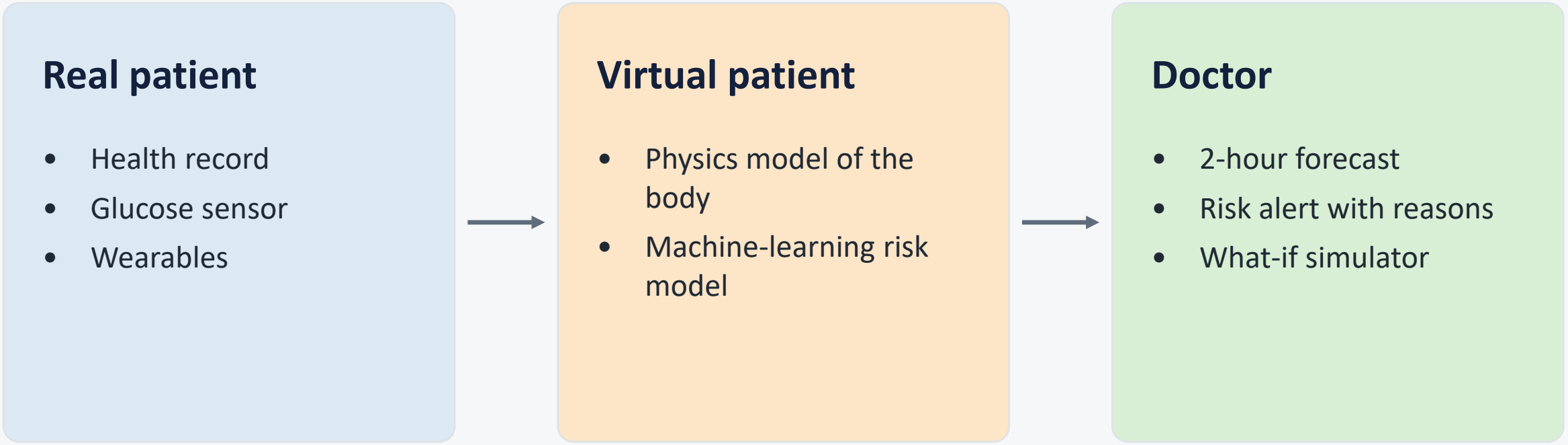
What will happen to this patient next, and what would change it?



	Projections (in millions)	95% CI
Diabetes	101·3	90·7–111·1
Prediabetes	136·0	123·5–147·5
Dysglycaemia	236·4	222·2–251·5
Hypertension	315·5	300·4–331·5
Generalised obesity	254·2	239·1–269·3
Abdominal obesity	351·1	335·1–367·9
Hypercholesterolaemia	213·3	179·5–247·1
High LDL cholesterol	185·7	153·8–217·7

https://www.thelancet.com/journals/landia/article/PIIS2213-8587(23)00119-5/fulltext

Our idea: a digital twin of the patient



A twin is more than a prediction: it can be experimented on, safely.

Scope: one condition, one question

Type 2 Diabetes

Will glucose leave 70 to 180 mg/dL in the next 2 hours, and why?

1

Forecast

2-hour glucose curve with an uncertainty band

2

Risk alert

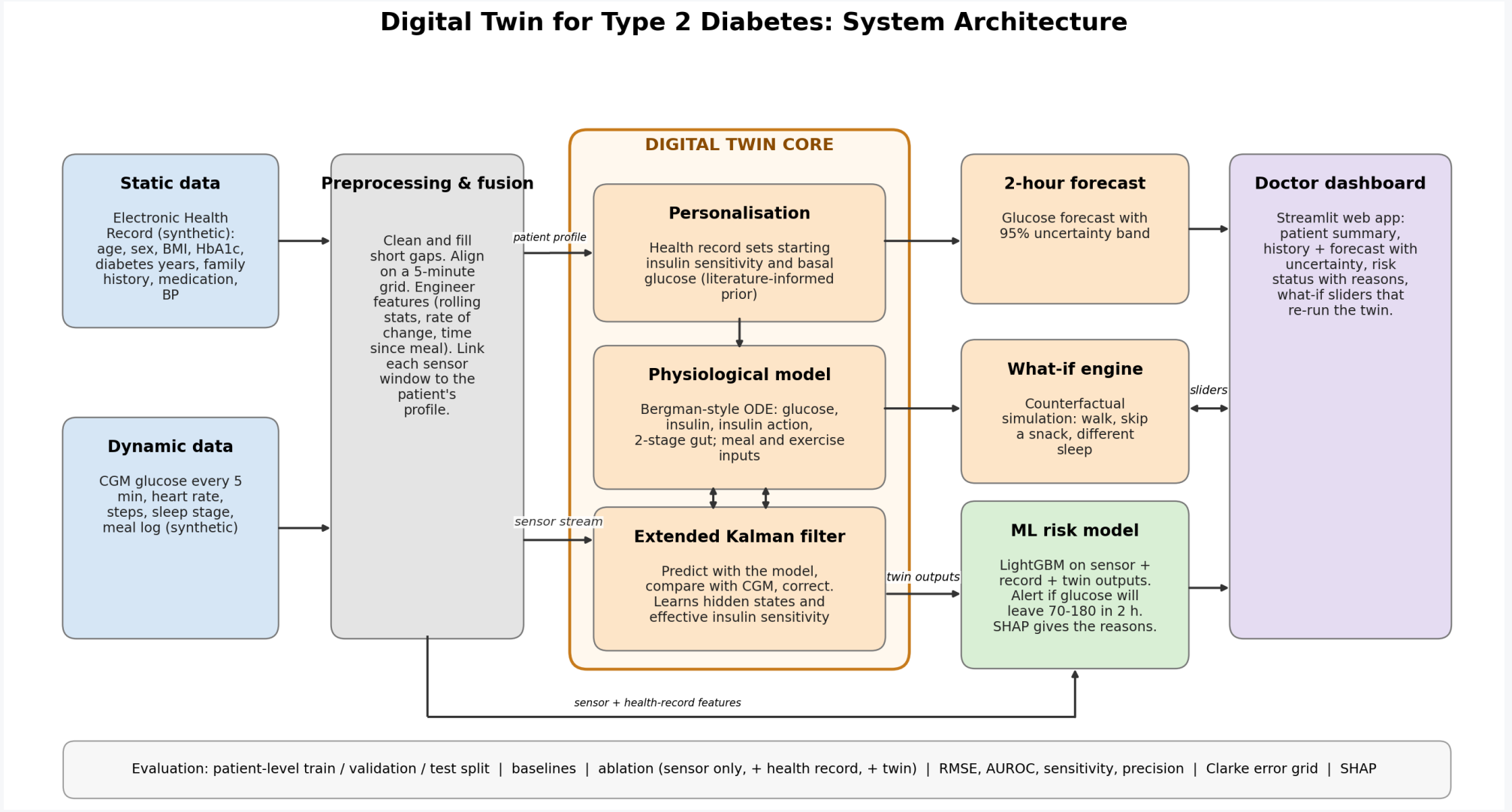
A score for going too high or too low, with the reasons

3

What-if

Change a meal, a walk or sleep and re-run the future

System architecture



Data: synthetic by design

Static: health record

- Age and sex
- BMI and HbA1c
- Years with diabetes
- Family history and metformin use
- Systolic blood pressure

Dynamic: sensor stream

- Glucose every 5 minutes
- Heart rate and steps
- Sleep stage
- Meal log (10% of meals unlogged)

60 patients × 14 days

All synthetic, no real person

Generator ≠ twin equations

Avoids testing the twin on its own maths

Glucose tied to HbA1c

Record and sensor tell one story

The physiological model: the twin’s engine

Meal → gut → blood

Carbs enter the stomach, then the intestine, then raise glucose about an hour later

Insulin → insulin action

Glucose triggers insulin, which builds an action that pulls glucose down

Exercise

A walk adds extra glucose uptake that lingers for a while

Insulin sensitivity

Per patient. Starts from the health record, then refined by the Kalman filter

$$\textit{Glucose change} = \textit{self-clearance} - \textit{insulin action} \times \textit{glucose} + \textit{meal absorption}$$

State estimation: the Kalman filter

1 Predict

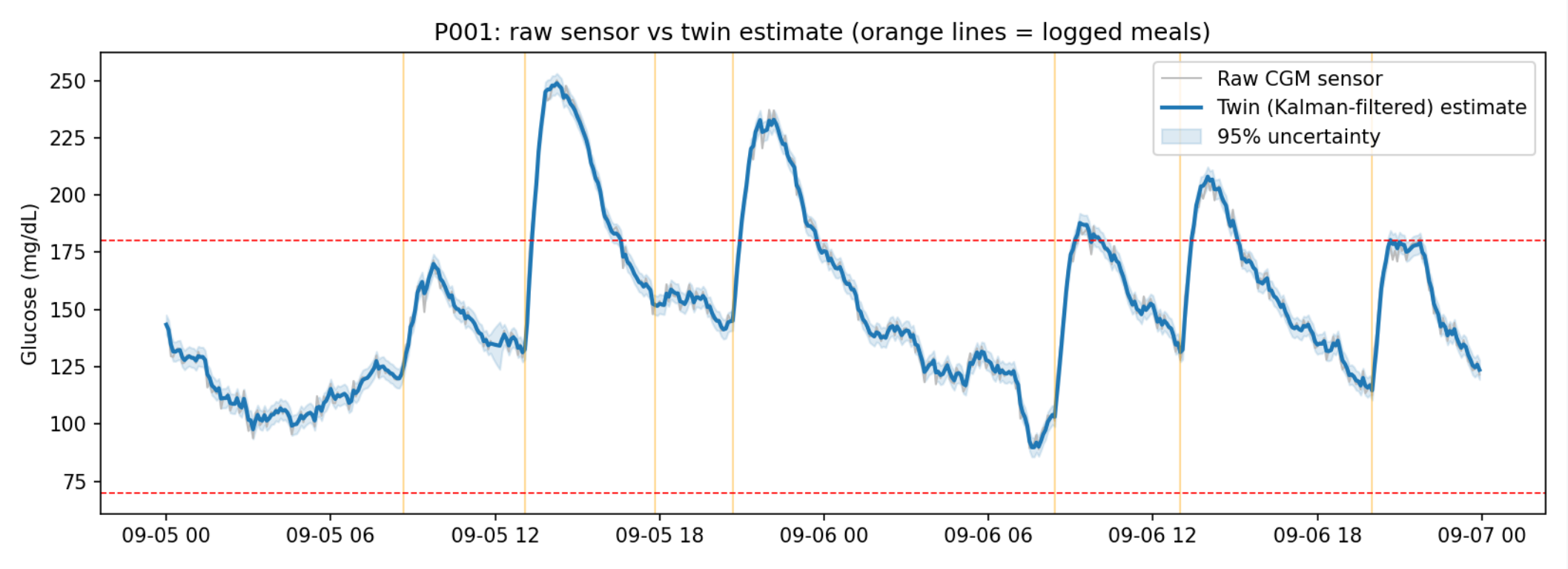
Model estimates the next 5 minutes

2 Compare

Read what the sensor reports

3 Correct

Blend by how reliable each is



Machine-learning risk model, and why it alerts

Sensor features

glucose, trend, meals, steps, sleep

Health record

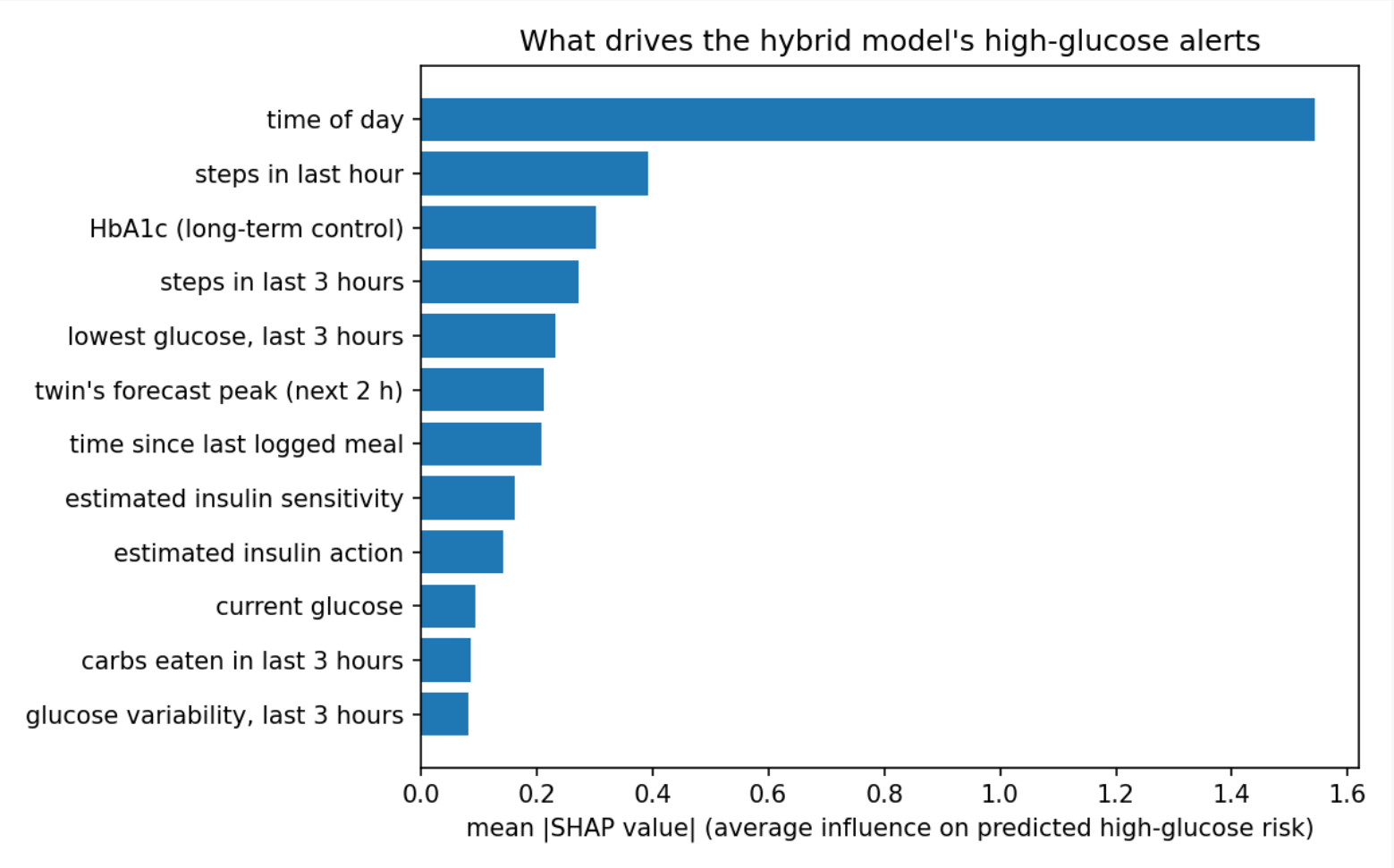
age, BMI, HbA1c, duration

Twin outputs

forecast peak, insulin action

LightGBM + SHAP

Output: risk score plus the top reasons



Results: will glucose go above 180 in the next 2 hours?

Model	AUROC	Sensitivity	Precision
Rule: linear extrapolation	0.63	0.69	0.21
Twin physics only	0.68	0.67	0.24
LightGBM: sensor only	0.91	0.64	0.64
LightGBM: + health record	0.92	0.69	0.63
LightGBM: + record + twin (hybrid)	**0.93**	**0.69**	**0.66**

Test set: 9 unseen patients, 4,707 moments, 800 high-glucose events. Alert threshold chosen on validation patients.

42 / 9 / 9

train / validation / test patients

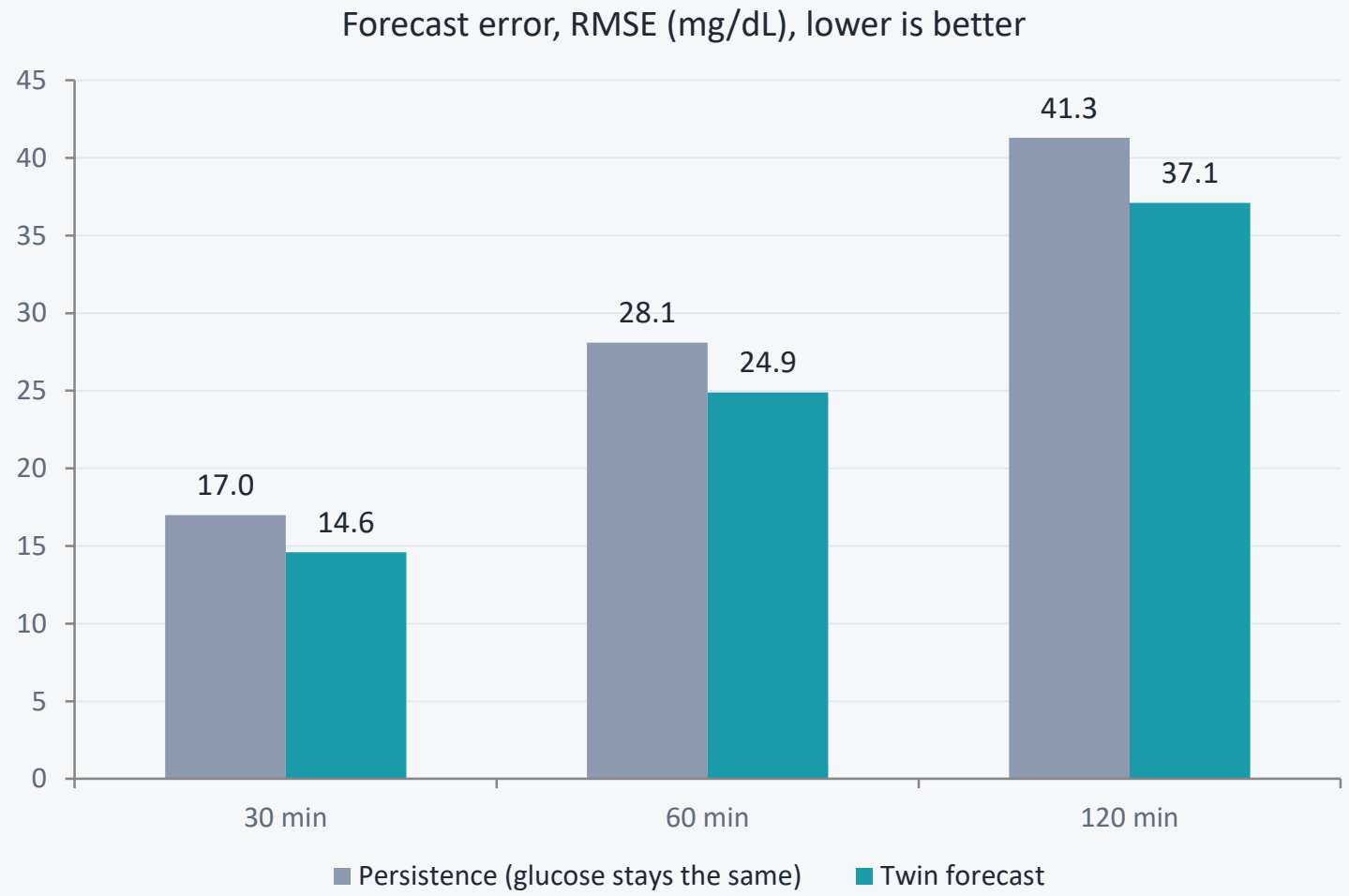
0.56

AUROC with shuffled labels: near chance, no leak

Small gaps

last two steps overlap in confidence intervals

Results: forecast accuracy



99%

of 1-hour forecasts in Clarke zones A+B (clinically safe)

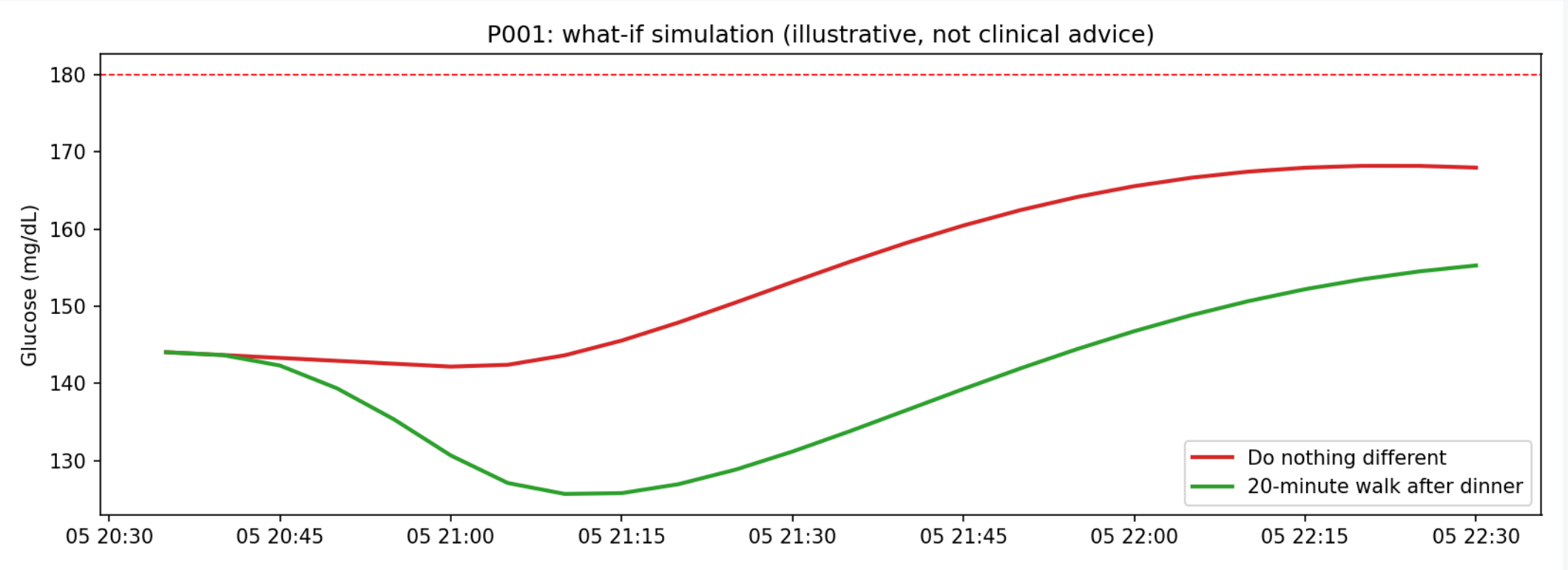
28.1 → 24.9

average error at 1 hour, baseline vs twin

No future meals

assumed unless the patient logs them

What-if: the twin as a simulator



Change the meal, the walk or last night's sleep, and re-run the next two hours on the virtual patient.

Illustrative simulation, not clinical advice. The sleep effect is an assumption.

Live prototype demo

1. Pick a virtual patient and a moment in time
2. Read the patient summary and glucose now
3. Forecast with uncertainty band; show what really happened
4. Risk status, score and the reasons (SHAP)
5. What-if: snack, walk, sleep: compare the curves
6. Switch patient; open the evaluation section

Screen recording of the dashboard
(cut here in the video)



Digital Twin_ Type 2 Diabetes - Personal - Microsoft? Edge 202

Limitations and ethics

- Synthetic data: no clinical accuracy claimed
- Simplified physiology: no stress, illness or other drugs
- Sleep and exercise effects are assumptions
- Only 9 test patients: small gaps are within noise

- Low-glucose prediction is exploratory
- Risk scores are not calibrated probabilities
- Forecast assumes no future meals unless logged
- Supports a doctor, never replaces one

Real use would need clinical validation, regulatory approval and DPDP-compliant privacy protection.

Future scope and tech stack

Next steps

- Validate on real, consented data with a hospital partner
- Connect to live wearables
- Hypertension and cardiovascular risk with the same pattern
- A patient-facing version with simple nudges

Tech stack

- Python, NumPy, pandas, SciPy
- LightGBM, scikit-learn, SHAP
- Streamlit and Plotly
- MIT licence, public GitHub repository

Thank you

A digital twin that fuses the health record with sensor data, forecasts glucose 2 hours ahead, explains why, and lets a doctor test changes first.

https://github.com/lucid-vini/Room209_USAR